Домашняя работа №1

№1.18

Дано:

$r(t)$

$\varphi(t)$

$\bar w \parallel \bar r$ − ?

$r^2\dot\varphi = const$

$r^2\dot\varphi = a$

Решение:

$\bar r(r,\varphi) = \begin{cases} r = r(t) \\ \varphi = \varphi(t) \end{cases}$ $\begin{cases} x = r\cos\varphi = r(t)\cos\varphi(t) \\ y = r\sin\varphi = r(t)\sin\varphi(t) \end{cases}$

$\bar v = (\dot x, \dot y) = (\dot r\cos\varphi - \sin\varphi\cdot r\cdot\dot\varphi,\ \dot r\sin\varphi + r\cos\varphi\cdot\dot\varphi)$

$\bar w = \dot{\bar v} = (\ddot x, \ddot y) = \cancel{(\ddot r\cos\varphi - \dot r\sin\varphi\cdot\dot\varphi - \cos\varphi}$

$\ddot x = \ddot r\cos\varphi - \dot r\sin\varphi\cdot\dot\varphi - (\cos\varphi\cdot\dot\varphi\cdot r\cdot\dot\varphi + (\dot r\dot\varphi + r\cdot\ddot\varphi)\sin\varphi)$

$\ddot y = \ddot r\sin\varphi + \dot r\cos\varphi\cdot\dot\varphi + \ddot\varphi r\cos\varphi + \dot\varphi(\dot r\cos\varphi - \dot\varphi\sin\varphi\, r)$

$$\begin{cases} \ddot x = \ddot r\cos\varphi - 2\sin\varphi\,\dot r\dot\varphi - r\cos\varphi\cdot\dot\varphi^2 - r\sin\varphi\cdot\ddot\varphi \\ \ddot y = \ddot r\sin\varphi + 2\cos\varphi\,\dot r\dot\varphi - r\sin\varphi\,\dot\varphi^2 + r\cos\varphi\,\ddot\varphi \end{cases}$$

$$[\bar r\times\bar w] = \begin{vmatrix} \bar e_1 & \bar e_2 & \bar e_3 \\ x & y & 0 \\ \ddot x & \ddot y & 0 \end{vmatrix} = (x\ddot y - y\ddot x)\bar e_3$$

$x\ddot y - y\ddot x = r\cos\varphi(\ddot r\sin\varphi + 2\cos\varphi\,\dot r\dot\varphi - r\sin\varphi\,\dot\varphi^2 + r\cos\varphi\,\ddot\varphi) - r\sin\varphi\cdot$
$\cdot(\ddot r\cos\varphi - 2\sin\varphi\,\dot r\dot\varphi - r\cos\varphi\,\dot\varphi^2 - r\sin\varphi\cdot\ddot\varphi) = r^2\cos^2\varphi\,\ddot\varphi + r^2\sin^2\varphi\,\ddot\varphi +$
$+ 2r\cos^2\varphi\,\dot r\dot\varphi + 2r\sin^2\varphi\,\dot r\dot\varphi = r^2\ddot\varphi + 2r\dot r\dot\varphi = \frac{a}{\dot\varphi}\cdot\ddot\varphi + 2\dot r\cdot r\cdot\frac{a}{r^2} =$
$= a\left(\frac{\ddot\varphi}{\dot\varphi} + 2\cdot\frac{\dot r}{r}\right) = a\cdot\left(-\frac{2\dot r}{r} + \frac{2\dot r}{r}\right) = a\cdot 0 = 0 \Rightarrow [\bar r\times\bar w] = 0 \Rightarrow \bar r \parallel \bar w$ ч.т.д.

$r^2\dot\varphi = a$, $2r\cdot\dot r\dot\varphi + \ddot\varphi r^2 = 0$; $2\dot r\dot\varphi + \ddot\varphi r = 0$; $\cancel{2\dot r\dot\varphi + \ddot\varphi r}$

$2\dot r\dot\varphi = -\ddot\varphi r$; $\frac{2\dot r}{r} = \frac{-\ddot\varphi}{\dot\varphi}$

№1.25

Дано:

$w = a(v\times r)$

$\rho_k(r, v) = ?$

Решение:

$\rho_k = \frac{v^2}{w_n} = \frac{v^2}{a|v\times r|}$

$\bar w = \bar w_\tau + \bar w_n$; $\bar v \perp \bar w \Rightarrow \bar w_\tau = 0 \Rightarrow$

$\Rightarrow \bar w = \bar w_n$

Ответ: $\frac{v^2}{a|v\times r|}$

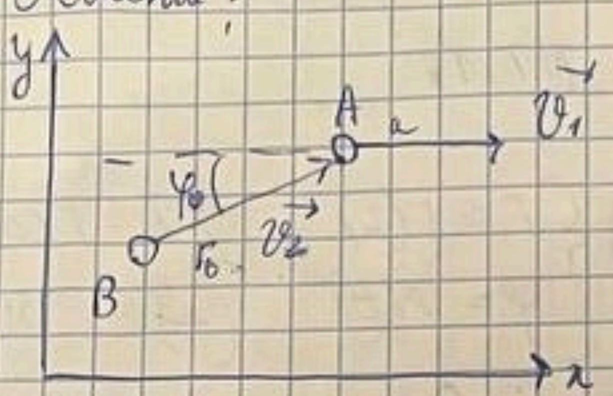
$$T_3 = \begin{pmatrix} f''(t)\sin\frac{\alpha_2}{2}\cdot\left(\frac{-v^2}{\ell\sin\frac{\alpha_2}{2}}\right) \\ f''(t)\cos\frac{\alpha_2}{2} \\ -\frac{2v}{\ell}f'(t) \end{pmatrix} = \begin{pmatrix} -f''(t)\frac{v^2}{\ell} \\ f''(t)\cos\frac{\alpha_2}{2} \\ -\frac{2v}{\ell}f'(t) \end{pmatrix}$$

$W_M^{abs} = \sqrt{f''(t)^2\frac{v^4}{\ell^2} + f''(t)^2\cos^2\frac{\alpha_2}{2} + \frac{4v^2}{\ell^2}f'(t)^2}$

Ответ: $v_M^{abs} = \sqrt{f'(t)^2 + \frac{v^2}{4}}$

Дано:
v_1, v_2
$AB = r(\varphi) = ?$
$\varphi \neq 0 \quad \widehat{a, BA}$

Решение: №1.31

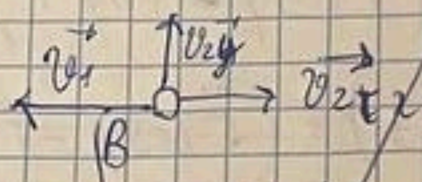


Начало отсчёта $A = (0,0)$

~~$y = -r_0 \sin\varphi_0 + v_2 \sin\varphi \cdot t$~~

~~$x = -r_0 \cos\varphi_0 + v_2 \cos\varphi t - v_1 t$~~

~~$r = r(x, y)$~~



$t = \frac{x + r_0 \cos\varphi_0}{v_2 \cos\varphi - v_1}$; $y = -r_0 \sin\varphi_0 + v_2 \sin\varphi \cdot \frac{x + r_0 \cos\varphi_0}{v_2 \cos\varphi - v_1}$

$y = r \sin\varphi$

$x = r \cos\varphi$

$r \sin\varphi = -r_0 \sin\varphi_0 + v_2 \sin\varphi \cdot \frac{r\cos\varphi + r_0 \cos\varphi_0}{v_2 \cos\varphi - v_1}$

$r\left(\sin\varphi - \frac{v_2 \sin\varphi \cos\varphi}{v_2 \cos\varphi - v_1}\right) = -r_0 \sin\varphi_0 + \frac{v_2 \sin\varphi \cdot r_0 \cos\varphi_0}{v_2 \cos\varphi - v_1}$

$r\left(\frac{v_2 \sin\varphi \cos\varphi - v_1 \sin\varphi - v_2 \sin\varphi \cos\varphi}{v_2 \cos\varphi - v_1}\right)$

~~$r(\varphi) =$~~

$r(\varphi) = \frac{+r_0 \sin\varphi_0 (v_2 \cos\varphi - v_1)}{v_1 \sin\varphi} - \frac{v_2 \sin\varphi \cos\varphi_0 r_0}{v_1 \sin\varphi}$

$r(\varphi) = \frac{r_0 (v_2 \cos\varphi - v_1) \sin\varphi_0}{v_1 \sin\varphi} - \frac{v_2 r_0 \cos\varphi_0}{v_1}$

~~$\frac{r(\varphi)}{r_0} = \frac{\sin\varphi_0}{\sin\varphi}\left(v_2 \cos\varphi / v_1 - 1 - \frac{v_2}{v_1} \cdot \sin\varphi \cdot \operatorname{ctg}\varphi_0\right)$~~

~~$\frac{r}{r_0} = \frac{\sin\varphi_0}{\sin\varphi} \cdot \frac{v_2}{v_1}\left(\cos\varphi - \frac{v_1}{v_2} - \frac{\sin\varphi}{\operatorname{tg}\varphi_0}\right)$~~ $= \frac{-\sin\varphi_0}{\sin\varphi}\left(1 - \frac{v_2}{v_1}\left(\cos\varphi - \frac{\sin\varphi}{\operatorname{tg}\varphi_0}\right)\right)$

$r(\varphi) = \frac{r_0 (v_2 \cos\varphi - v_1) \sin\varphi_0 - v_2 r_0 \cos\varphi_0 \sin\varphi}{v_1 \sin\varphi}$

Ответ: $\frac{\rho}{\rho_0} = \frac{\sin\varphi_0}{\sin\varphi} \cdot \left(\frac{\operatorname{tg}\varphi/2}{\operatorname{tg}\varphi_0/2}\right)^{v_2/v_1}$

$\int \frac{1}{\sin x} dx = -\int \frac{d\cos x}{1 - \cos^2 x} = \frac{1}{2} \ln\left|\frac{\cos x - 1}{\cos x + 1}\right| = \ln\left|\operatorname{tg}\frac{x}{2}\right|$

$\int \operatorname{ctg} x\, dx = \int \frac{d\sin x}{\sin x} = \ln|\sin x|$

$\frac{r(\varphi)}{r_0} = \frac{v_2 \cos\varphi \sin\varphi_0 - v_2 \cos\varphi_0 \sin\varphi}{v_1 \sin\varphi} - \frac{\sin\varphi_0}{\sin\varphi}$

$-S \sin\varphi + \cos\varphi = -\cos\varphi + \frac{v_1}{v_2} + \cos\varphi = \frac{v_1}{v_2}$

~~$x = r\cos\varphi \cos(90 - \alpha)$~~

~~$y = r\cos\varphi \sin(90 - \alpha)$~~

~~$z = r\sin\varphi$~~

$\begin{cases} x = \rho\cos\varphi \\ y = \rho\sin\varphi \end{cases}$ $\begin{vmatrix} dx = d\rho\cos\varphi - \rho\sin\varphi d\varphi \\ dy = d\rho\sin\varphi + \rho\cos\varphi d\varphi \end{vmatrix}$

$\begin{cases} dx = -v_1 dt + v_2 \cos\varphi\, dt \\ dy = v_2 \sin\varphi\, dt \end{cases}$

$S = \operatorname{ctg}\varphi - \frac{v_1}{v_2 \sin\varphi} = \frac{d\rho\cos\varphi - \rho\sin\varphi d\varphi}{d\rho\sin\varphi + \rho\cos\varphi d\varphi}$; $S(d\rho\sin\varphi + \rho\cos\varphi d\varphi) = d\rho\cos\varphi - \rho\sin\varphi d\varphi$

$d\rho(S\sin\varphi - \cos\varphi) = -\rho d\varphi(S\cos\varphi + \sin\varphi)$

$\frac{d\rho}{\rho} = d\varphi \cdot \frac{S\cos\varphi + \sin\varphi}{-S\sin\varphi + \cos\varphi} = d\varphi \cdot \frac{v_2}{v_1}(S\cos\varphi + \sin\varphi) = \frac{v_2}{v_1} d\varphi \cdot \left(\frac{\cos^2\varphi}{\sin\varphi} - \frac{v_1}{v_2}\operatorname{ctg}\varphi + \sin\varphi\right) =$

$= \frac{v_2}{v_1} d\varphi \left(\frac{1}{\sin\varphi} - \frac{v_1}{v_2}\operatorname{ctg}\varphi\right) = \frac{v_2}{v_1 \sin\varphi} d\varphi - \operatorname{ctg}\varphi d\varphi$; $\ln\left|\frac{\rho}{\rho_0}\right| = \frac{v_2}{v_1} \ln\left|\frac{\operatorname{tg}\varphi/2}{\operatorname{tg}\varphi_0/2}\right| + \ln\left|\frac{\sin\varphi_0}{\sin\varphi}\right|$

№ 1

Дано:

r, λ, φ

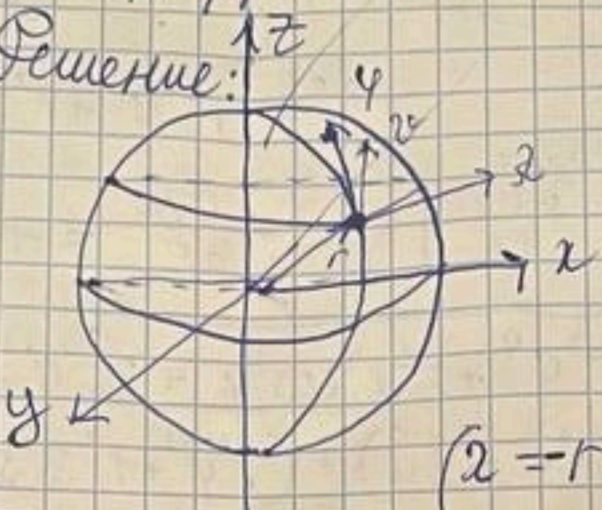
$\alpha; |\vartheta| = const$

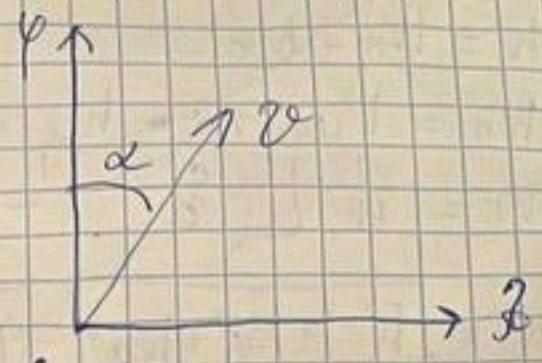
$W_r, W_\lambda, W_\varphi$ — ?

$|W| = ?$

$\rho_\kappa = ?$

Решение:





$$\bar r = \bar r(x, y, z) \qquad \bar r(\rho, \varphi, \lambda) \qquad \begin{cases} x = r \cos\varphi \cos\lambda \\ y = r \cos\varphi \sin\lambda \\ z = r \sin\varphi \end{cases}$$

$r = const \Rightarrow |\vartheta_r| = 0, \ |W_r| = 0$

$$H_\varphi = \sqrt{\left(\frac{\partial r_x}{\partial\varphi}\right)^2 + \left(\frac{\partial r_y}{\partial\varphi}\right)^2 + \left(\frac{\partial r_z}{\partial\varphi}\right)^2} = \sqrt{r^2 \sin^2\varphi \cos^2\lambda + r^2 \sin^2\varphi \sin^2\lambda + r^2 \cos^2\varphi} = r$$

$$H_\lambda = \sqrt{\left(\frac{\partial r_x}{\partial\lambda}\right)^2 + \left(\frac{\partial r_y}{\partial\lambda}\right)^2 + \left(\frac{\partial r_z}{\partial\lambda}\right)^2} = \sqrt{r^2 \cos^2\varphi \sin^2\lambda + r^2 \cos^2\varphi \cos^2\lambda + 0} = r\cos\varphi; \quad H_r = \sqrt{\left(\frac{\partial r_x}{\partial r}\right)^2 + \left(\frac{\partial r_y}{\partial r}\right)^2 + \left(\frac{\partial r_z}{\partial r}\right)^2} = 1$$

$$\vartheta_\varphi = H_\varphi \cdot \dot\varphi = r \cdot \dot\varphi, \quad \vartheta_\lambda = H_\lambda \dot\lambda = r \cdot \dot\lambda \cos\varphi, \quad \vartheta_r = H_r \cdot r' = 0$$

$$\vartheta = \sqrt{\vartheta_\varphi^2 + \vartheta_\lambda^2} \Rightarrow r^2 \dot\varphi^2 + r^2 \dot\lambda^2 \cos^2\varphi = \vartheta^2$$

$$\vartheta^2 = r^2 \dot\varphi^2 + r^2 \dot\lambda^2 \cos^2\varphi$$

$$W_\varphi = \frac{1}{H_\varphi} \cdot \left[\left(\frac{\partial \vartheta^2/2}{\partial\dot\varphi}\right)' - \frac{\partial \vartheta^2/2}{\partial\varphi}\right] = \frac{1}{r} \cdot \left[(r^2\dot\varphi)' + r^2 \dot\lambda^2 \sin\varphi\cos\varphi\right] =$$

$$= r\ddot\varphi + r\dot\lambda^2 \sin\varphi\cos\varphi =$$

$$W_\lambda = \frac{1}{H_\lambda}\left[\left(\frac{\partial \vartheta^2/2}{\partial\dot\lambda}\right)' - \frac{\partial \vartheta^2/2}{\partial\lambda}\right] = \frac{1}{r\cos\varphi} \cdot \left[\left(r^2 \dot\lambda \cos^2\varphi\right)'\right] =$$

$$= \frac{1}{r\cos\varphi} \cdot \left(r^2 \ddot\lambda \cos^2\varphi - 2r^2 \dot\lambda \sin\varphi\cos\varphi \cdot \dot\varphi\right) = r\ddot\lambda \sin\varphi \ + 2r\dot\lambda \sin\varphi \cdot \dot\varphi =$$

$= \frac{1}{r}v^2 \sin\alpha \cancel{\operatorname{ctg}\varphi \cdot \cos\alpha (2 - \sin\varphi)} = \frac{v^2}{r} \sin\alpha\cos\alpha (2\operatorname{ctg}\varphi - \cos\varphi)$

$\cancel{W = W_n + W_\tau}$

$\cancel{W_n = W_\varphi \sin\alpha + W_\lambda \cos\alpha =}$

$v^2 = r^2\dot\varphi^2 + r^2\dot\lambda^2\cos^2\varphi + \dot r^2$

$W_r = \frac{1}{4r}\left[\left(\frac{\partial v^2/2}{\partial \dot r}\right)^{\cdot} - \frac{\partial v^2/2}{\partial r}\right] = -\frac{\partial v^2/2}{\partial r} = -r\dot\varphi^2 - r\dot\lambda^2\cos^2\varphi$

$|W| = \sqrt{W_r^2 + W_\varphi^2 + W_\lambda^2} = \sqrt{(r^2\dot\varphi^4 + r^2\dot\lambda^4\cos^4\varphi + 2r^2\dot\varphi^2\dot\lambda^2\cos^2\varphi) + (r^2\ddot\lambda^2\sin^2\varphi + 4r^2\dot\lambda^2\sin^2\varphi\,\dot\varphi^2 + 4r^2\sin^2\varphi\,\ddot\lambda\dot\lambda\dot\varphi) + (r^2\ddot\varphi^2 + r^2\dot\lambda^4\sin^2\varphi\cos^2\varphi + 2r^2\cos\varphi\sin\varphi\,\ddot\varphi\dot\lambda^2)}$

$\rho = \frac{-v^2}{W_r} = \frac{-v^2}{r\dot\varphi^2 + r\dot\lambda^2\cos^2\varphi}$

N3.2

Дано:

$v_A \neq v_B$, $\frac{v_A}{v_B} = \lambda$

$\rho(A, B) = a$

ϱ ГМТ?

Решение:

Пусть $A = (0,0)$, $B = (0, a)$; $C(x, y)$

$\bar v_A = \bar v_B + [\bar\omega \cdot \bar{BA}]$; $v_A \begin{pmatrix} v_{xA} \\ v_{yA} \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xB} \\ v_{yB} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \omega \end{pmatrix} \times \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xB} + a\omega \\ v_{yB} \\ 0 \end{pmatrix}$

$\bar v_C = \bar v_B + [\bar\omega_B \times \bar{BC}]$

$\bar v_C = \bar v_A + [\bar\omega_A \times \bar{AC}]$

C – мгн. центр скорост.

$\Rightarrow \bar\omega_B = \bar\omega_A$; $\bar v_C = 0$

$= \omega' = \omega$

$\begin{pmatrix} v_{xC} \\ v_{yC} \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xB} \\ v_{yB} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \omega_B \end{pmatrix} \times \begin{pmatrix} x \\ y - a \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xB} + (a - y)\omega_B \\ v_{yB} + x\omega_B \\ 0 \end{pmatrix}$

$\begin{pmatrix} v_{xC} \\ v_{yC} \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xA} \\ v_{yA} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \omega_B \end{pmatrix} \times \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} v_{xA} - \omega_B y \\ v_{yA} + \omega_B x \\ 0 \end{pmatrix}$

$-v_{xB} + v_{xA} = \omega \cdot a$

$v_{yB} - v_{yA} = 0$, $v_{yB} = v_{yA} = v_y$; $v_{xA}^2 + v_y^2 = \lambda^2 v_{xB}^2 + \lambda^2 v_y^2$

$$\begin{cases} v_{xA}^2 = \lambda^2 v_{xB}^2 + (\lambda^2 - 1)v_y^2 = v_{xB}^2 + \omega^2 a^2 + 2v_{xB}\omega a \\ \quad (\lambda^2 - 1)(v_{xB}^2 + v_y^2) = \omega a(\omega a + 2v_{xB}) \\ v_{xA} = v_{xB} + \omega a \\ v_{xB} + (a - y)\omega = 0, \; y = \frac{v_{xB}}{\omega} + a, \; (y - a)^2 = \frac{v_{xB}^2}{\omega^2} \\ v_y + \omega x = 0; \; x = -\frac{v_y}{\omega}, \; x^2 = \frac{v_y^2}{\omega^2} \end{cases}$$

$((y - a)^2 + x^2)\omega^2 = v_{xB}^2 + v_y^2$

$(\lambda^2 - 1)((y - a)^2 + x^2) = a\left(a + \frac{2v_{xB}}{\omega}\right) = a(a + 2(y - a)) = a(2y - a)$

$\cancel{y^2+2ay+a^2+\varkappa^2}$

$x^2+(y+a)^2 = \cancel{\frac{a(a+2y)}{\varkappa^2-1}}\ \frac{a(2y+a)}{1-\varkappa^2} = \frac{a^2}{1-\varkappa^2}+\frac{2ay}{1-\varkappa^2}$

$(\varkappa^2-1)(x^2+(y+a)^2) = -2ay-a^2$

$\varkappa^2x^2+\varkappa^2(y+a)^2-x^2-y^2+2ay-a^2=-a^2$

$\varkappa^2x^2+\varkappa^2(y+a)^2 = x^2+y^2$

$\varkappa^2(x^2+(y-a)^2) = x^2+y^2$; $\varkappa^2x^2-x^2+(y^2-2ay+a^2)\varkappa^2-y^2=0$

~~Ответ:~~ $\cancel{\varkappa^2(x^2-1)+y\varkappa^2}$ $x^2(\varkappa^2-1)+y^2(\varkappa^2-1)-2ay\varkappa^2=-a^2\varkappa^2$

$\cancel{(\varkappa^2-1)(x^2+y^2)=\varkappa^2}$ $x^2(\varkappa^2-1)+y^2(\varkappa^2-1)$

$x^2+y^2-2y\cdot\frac{a\varkappa^2}{\varkappa^2-1} = -\frac{a^2\varkappa^2}{\varkappa^2-1}$

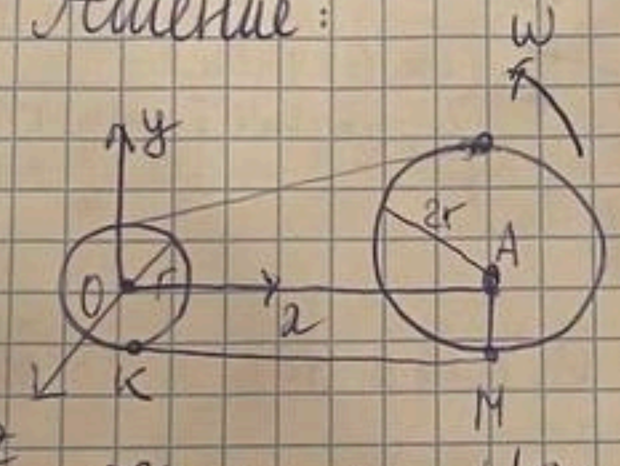
$x^2+\left(y-\frac{a\varkappa^2}{\varkappa^2-1}\right)^2 = \frac{a^2\varkappa^4}{(\varkappa^2-1)^2}-\frac{a^2\varkappa^2}{\varkappa^2-1} = a^2\left(\frac{\varkappa^4}{\varkappa^2-1}-\frac{\varkappa^2(\varkappa^2-1)}{(\varkappa^2-1)^2}\right) = \frac{a^2\varkappa^2}{(\varkappa^2-1)^2}$

Ответ: $x+\left(y-\frac{a\varkappa^2}{\varkappa^2-1}\right)^2 = \frac{a^2\varkappa^2}{(\varkappa^2-1)^2}$

N3.20

Дано:
l, r
$R=2r$
ω, ε
$v_M=?$
$W_M=?$

Решение:



$\bar v_M = \bar v_A + \bar\omega_M \times \overline{AM}$

$\bar v_A = \bar v_O^{\,=0} + \bar\omega \times \overline{OA}$

$\bar v_A = \begin{pmatrix}0\\0\\\omega\end{pmatrix}\times\begin{pmatrix}l\\0\\0\end{pmatrix} = \begin{pmatrix}0\\\omega l\\0\end{pmatrix}$

$\cancel{|v_K| = |v_M|}$ $|v_{отн\,M}| = |v_K|$, условие зацепления

$-\omega_M\cdot 2r = \omega r$, $\omega_M = -\frac{\omega}{2}$

$\bar v_M = \begin{pmatrix}0\\\omega l\\0\end{pmatrix}+\begin{pmatrix}0\\0\\-\frac{\omega}{2}\end{pmatrix}\times\begin{pmatrix}0\\-2r\\0\end{pmatrix} = \begin{pmatrix}0\\\omega l\\0\end{pmatrix}+\begin{pmatrix}-\omega r\\0\\0\end{pmatrix} = \begin{pmatrix}-\omega r\\\omega l\\0\end{pmatrix}$

$|\bar v_M| = \omega\sqrt{l^2+r^2}$

$\bar W_M = \bar W_A + \bar\omega_M\times[\bar\omega_M\times \overline{AM}] + \bar\varepsilon_M\times\overline{AM} = \begin{pmatrix}-\omega^2 l\\\varepsilon l\\0\end{pmatrix}+\begin{pmatrix}\frac{\omega^2 l}{2}\\-\frac{\omega^2 r}{2}\\0\end{pmatrix}+\begin{pmatrix}-\varepsilon r\\0\\0\end{pmatrix} = \begin{pmatrix}-\frac{\omega^2 l}{2}-\varepsilon r\\\varepsilon l-\frac{\omega^2 r}{2}\\0\end{pmatrix}$

$\bar W_A = \cancel{\bar W_O}^{\,0} + \bar\omega\times[\bar\omega\times \overline{OA}] + \bar\varepsilon\times \overline{OA} = \bar\omega\times\bar v_A + \bar\varepsilon\times\overline{OA} =$

$= \begin{pmatrix}0\\0\\\omega\end{pmatrix}\times\begin{pmatrix}0\\\omega l\\0\end{pmatrix}+\begin{pmatrix}0\\0\\\varepsilon\end{pmatrix}\times\begin{pmatrix}l\\0\\0\end{pmatrix} = \begin{pmatrix}-\omega^2 l\\+\varepsilon l\\0\end{pmatrix}$

$|W_M| = \sqrt{\frac{\omega^4 l^2}{4}+\frac{\omega^4 r^2}{4}+\varepsilon^2(r^2+l^2)} = \sqrt{\left(\varepsilon^2+\frac{\omega^4}{4}\right)(l^2+r^2)}$

$\bar\omega_M\times[\bar\omega_M\times\overline{AM}] = \bar\omega_M\times\bar v_M = \begin{pmatrix}0\\0\\-\frac{\omega}{2}\end{pmatrix}\times\begin{pmatrix}-\omega r\\\omega l\\0\end{pmatrix} = \begin{pmatrix}\frac{\omega^2 l}{2}\\\frac{\omega^2 r}{2}\\0\end{pmatrix}$

$\bar\varepsilon_M = -\frac{\varepsilon}{2}$, $\bar\varepsilon_M\times\overline{AM} = \begin{pmatrix}0\\0\\-\frac{\varepsilon}{2}\end{pmatrix}\times\begin{pmatrix}0\\-2r\\0\end{pmatrix} = \begin{pmatrix}-\varepsilon r\\0\\0\end{pmatrix}$

Ответ:

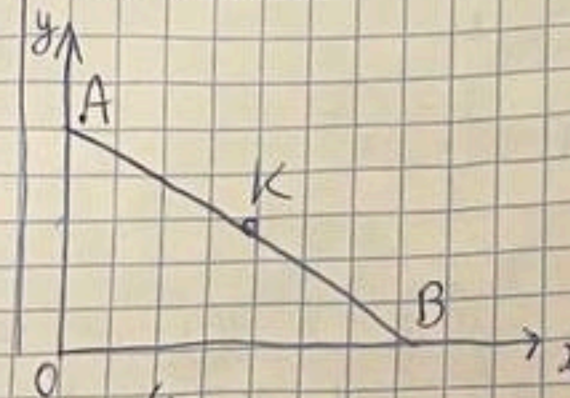
$|\bar v_M| = \omega\sqrt{l^2+r^2}$

$|W_M| = \cancel{\sqrt{\frac{\omega^4 l^2}{4}+\frac{\omega^4 r^2}{4}}}$ $\sqrt{\left(\varepsilon^2+\frac{\omega^4}{4}\right)(l^2+r^2)}$

N3.21

Дано:
$v_A = const$
$\overline{W_K} \perp Oy$
$|W_K| \sim \frac{1}{S^3}$

Решение:



K - произвольная точка на AB

$\cancel{\overline{v_K} = \overline{v_A} + \overline{\omega} \times \overline{AK}}$

$y_A^2 + x_B^2 = const = L^2$

$x_B = \sqrt{L^2 - y_A^2}$

$\cancel{y_A = (y_0 - v_A t)}$, $\cancel{x_B = \sqrt{L^2 - (y_0 - v_A^2 t^2)}}$

$\cancel{\dot{x}_B = \frac{-2 v_A^2 t}{2\sqrt{L^2 - v_A^2 t^2}} = -\frac{v_A^2 t}{\sqrt{L^2 - v_A^2 t^2}}}$

$y_A = y_0 - v_a t$, $x_B = \sqrt{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_A t}$

$\dot{x}_B = \frac{-v_a^2 t + y_0 v_a}{\sqrt{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_a t}}$

$\ddot{x}_B = \frac{-v_a^2 \sqrt{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_A t} + (v_a^2 t - y_0 v_A) \cdot \frac{-v_a^2 t + y_0 v_a}{\sqrt{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_a t}}}{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_a t} =$

$\cancel{= \frac{-v_a^2}{\sqrt{L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_a t}} - \frac{(v_a^2 t + y_0 v_a)^2}{(L^2 - y_0^2 - v_a^2 t^2 + 2 y_0 v_a t)^{3/2}} =}$

$\cancel{= -\frac{v_a^2}{\sqrt{x_B}} - \frac{(v_a t - y_0 v_a)^2}{\sqrt{x_B^3}}}$

$= \frac{-v_a^2 L^2 + v_a^2 (v_a t - y_0)^2 - v_a^2 (v_a t - y_0)}{x_B^2} = \frac{-v_a^2 L^2}{x_B^2}$

Ускорение произвол. точки K $\ddot{x}_K = \ddot{x}_B \cdot \frac{d}{x_B}$,

$\ddot{x}_K \cdot x_B / d = \frac{-v_a^2 L^2}{x_B^2}$; $\ddot{x}_K = \frac{-v_a^2 L^2 d}{x_B^3}$, $x_B \sim x_K \Rightarrow \ddot{x}_K \sim \frac{-v_a^2 L^2 d}{x_K^3}$ ч.т.д.

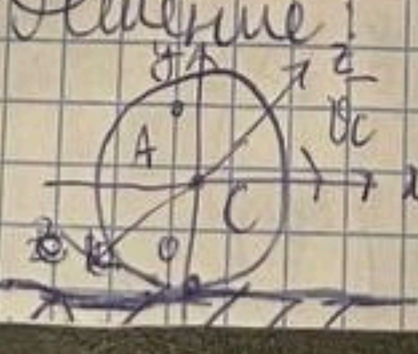
Ускорение любой точки перпендикулярно Oy, т.к. $\ddot{y} = 0$. А ускорение по y равно 0, т.к. $W_a = 0$ (иначе стержень в каких-то точках начал растягиваться $\Rightarrow$ $\Rightarrow$ изменилась бы длина - что невозм. для ТТ).

Ответ: $\ddot{x}_K \sim \frac{-v_a^2 L^2 d}{x_K^3}$

N3.25 $\odot \overline{\omega}$

Дано:
R, v_c, W_c, $A(x, y)$
$W_t = ?$, $W_n = ?$

Решение:



$v_0 = 0$, т.к. без проскальз.

$\overline{v_C} = \begin{pmatrix} v_C \\ 0 \\ 0 \end{pmatrix} +$

$\bar{v}_C = \bar{v}_O + \bar{\omega} \times \overline{OC}$, $\begin{pmatrix} v_c \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ +\omega \end{pmatrix} \times \begin{pmatrix} 0 \\ R \\ 0 \end{pmatrix} = \begin{pmatrix} -\omega R \\ 0 \\ 0 \end{pmatrix}$

$|\omega| = \frac{v_c}{R}$. Пусть $C = (0, R)$, $v_C = v$

$\omega = \begin{pmatrix} 0 \\ 0 \\ -v/R \end{pmatrix}$

$\bar{v}_A = \bar{v}_c + \bar{\omega} \times \overline{CA} = \begin{pmatrix} v_c \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -\frac{v_c}{R} \end{pmatrix} \times \begin{pmatrix} x \\ y-R \\ 0 \end{pmatrix} = \begin{pmatrix} v \cdot \frac{y}{R} \\ -\frac{v}{R} x \\ 0 \end{pmatrix}$

$\cancel{W_\tau = \frac{v_a^2}{\rho}}$, $v_a^2 = \frac{v^2}{R^2}(x^2+y^2)$, $\bar{\varepsilon} = \frac{d\bar{\omega}}{dt} = \begin{pmatrix} 0 \\ 0 \\ -\frac{W_c}{R} \end{pmatrix}$

$\bar{W}_A = \bar{W}_C + \bar{\omega} \times [\bar{\omega} \times \overline{CA}] + \bar{\varepsilon} \times \overline{CA} =$

$= \begin{pmatrix} W_c \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -v/R \end{pmatrix} \times \begin{pmatrix} v/R\,(y-R) \\ -v/R\, x \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -W_c/R \end{pmatrix} \times \begin{pmatrix} x \\ y-R \\ 0 \end{pmatrix} = \begin{pmatrix} W_c \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{v^2}{R^2} x \\ -\frac{v^2}{R^2}(y-R) \\ 0 \end{pmatrix} + \begin{pmatrix} -W_c + \frac{W_c}{R} y \\ -\frac{W_c}{R} x \\ 0 \end{pmatrix} =$

$= \begin{pmatrix} \frac{W_c R y - v^2 x}{R^2} \\ \frac{-v^2(y-R) - W_c x R}{R^2} \\ 0 \end{pmatrix}$, $\frac{\bar{v}_a}{|v_a|} = \frac{R}{v\sqrt{x^2+y^2}} \cdot \begin{pmatrix} v/R \cdot y \\ -v/R\, x \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{y}{\sqrt{x^2+y^2}} \\ \frac{-x}{\sqrt{x^2+y^2}} \\ 0 \end{pmatrix}$, $\frac{\bar{r}}{|r|} = \begin{pmatrix} \frac{x}{\sqrt{x^2+y^2}} \\ \frac{y}{\sqrt{y^2+x^2}} \\ 0 \end{pmatrix}$

$\bar{W}_A = \bar{W}_\tau + \bar{W}_n = |W_\tau| \cdot \bar{e}_\tau + |W_n| \cdot \bar{e}_n$, $\bar{e}_\tau = \frac{\bar{v}}{|v|}$, $\bar{e}_n = \frac{\bar{r}}{|r|}$

$|\bar{W}_\tau| = (W_A ; e_\tau) = \left(\bar{W}_A \cdot \frac{\bar{v}}{|v|}\right) = \frac{W_c R y^2 - v^2 x y}{R^2 \sqrt{x^2+y^2}} + \frac{v^2 (y-R) x + W_c x^2 R}{R^2 \sqrt{x^2+y^2}} =$

$= \frac{W_c}{R}\sqrt{x^2+y^2} - \frac{v^2 x}{R\sqrt{x^2+y^2}}$

$|W_n| = \left(\bar{W}_A, \frac{\bar{r}}{|r|}\right) = \frac{W_c R y x - v^2 x^2}{R^2\sqrt{x^2+y^2}} - \frac{v^2 (y-R) y + W_c x y R}{R^2 \sqrt{x^2+y^2}} =$

$= -\frac{v^2}{R^2}\sqrt{x^2+y^2} + \frac{v^2}{R} \cdot \frac{y}{\sqrt{x^2+y^2}}$

Ответ: $W_\tau = \frac{W_c}{R}\sqrt{x^2+y^2} - \frac{v^2 x}{R\sqrt{x^2+y^2}}$; $W_n = \frac{v^2}{R} \cdot \frac{y}{\sqrt{x^2+y^2}} - \frac{v^2}{R^2}\sqrt{x^2+y^2}$

№4.4

Ответ: $\omega = \sqrt{\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\cos\theta}$; $-\omega_1\omega_2\sin\theta = \varepsilon$

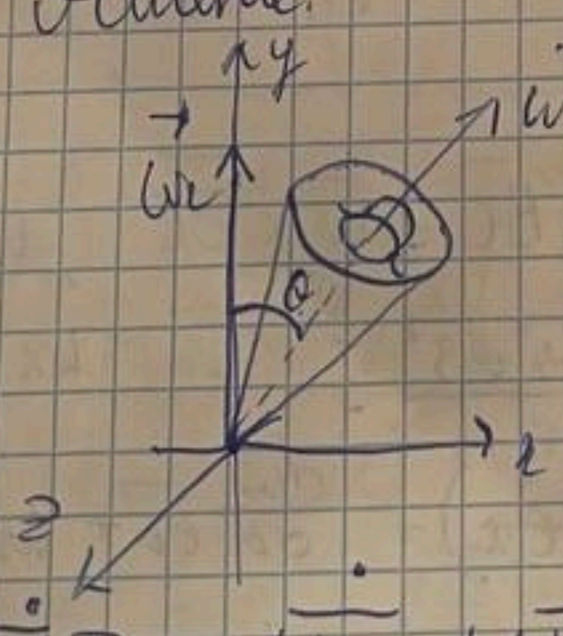
Дано:

ω_1, ω_2

$\omega = ?$

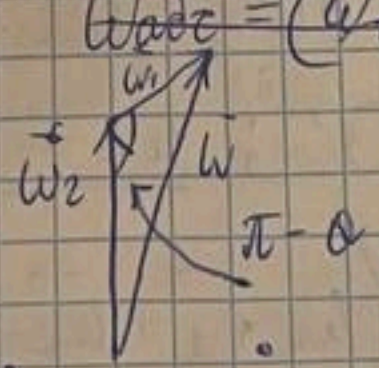
$\varepsilon = ?$

Решение:



1) $\bar{\omega}_{абс} = \bar{\omega}_1 + \bar{\omega}_2 = \bar{\omega}_{отн} + \bar{\omega}_{пер}$

$\cancel{\omega_{абс} = (\omega_1 \cdot \omega_2) = \begin{pmatrix} \omega_1 \sin\theta \\ \omega_2 \cos\theta \\ 0 \end{pmatrix} \cdot (}$

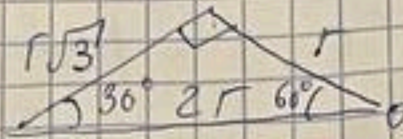


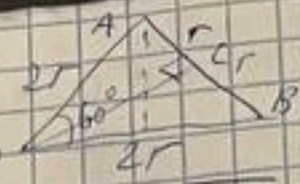
$\omega = \sqrt{\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\cos\theta}$

2) $\bar{\varepsilon}_{абс} = \dot{\bar{\omega}}_{абс} = \dot{\bar{\omega}}_{отн} + \dot{\bar{\omega}}_{пер} = \dot{\bar{\omega}}_1 + \underbrace{\dot{\bar{\omega}}_2}_{=0} = \dot{\bar{\omega}}_1 = \omega_1 \dot{\bar{e}}_1 = \cancel{\omega_1} \bar{\omega}_2$

$= \bar{\omega}_2 \cdot \omega_1 \cdot \bar{e}_1 = \bar{\omega}_2 \cdot \bar{\omega}_1 \Rightarrow \varepsilon_{абс} = \begin{pmatrix} \omega_1 \sin\theta \\ \omega_2 \cos\theta \end{pmatrix} \times \begin{pmatrix} 0 \\ \omega_2 \end{pmatrix} = -\omega_1\omega_2\sin\theta$

№4.10





Дано:

r;

$a = l = r\sqrt{3}$

$v_B = 0$

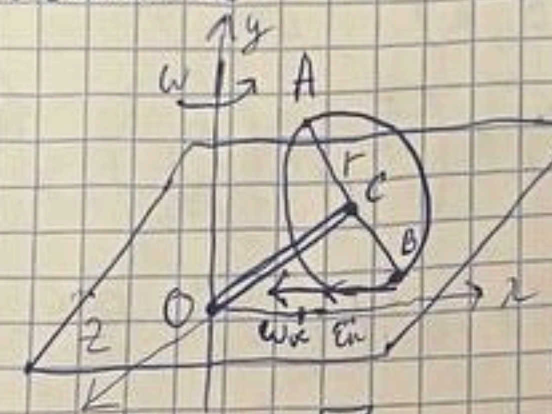
ω, ε

ω_K, ε_K — ?

~~$\varepsilon_A, \varepsilon_B$~~

w_A, w_B — ?

Решение:



$$\bar v_A = \bar v_C + \bar\omega_{отн} \times \overline{CA}$$

$$\bar v_C = \bar v_O + \bar\omega \times \overline{OC}$$

$$\overline{OC} = \left(\tfrac{3}{2}r,\ r\tfrac{\sqrt{3}}{2},\ 0\right)$$

$$\overline{CA} = (r,\ r\sqrt{3},\ 0) - \left(\tfrac{3}{2}r,\ r\tfrac{\sqrt{3}}{2},\ 0\right) = \left(-\tfrac{1}{2}r,\ r\tfrac{\sqrt{3}}{2},\ 0\right),\quad \overline{CB} = (2r,0,0) - \left(\tfrac{3}{2}r,\ r\tfrac{\sqrt{3}}{2},\ 0\right) = \left(\tfrac{1}{2}r,\ -r\tfrac{\sqrt{3}}{2},\ 0\right)$$

$$\bar v_C = \begin{pmatrix}0\\ \omega\\ 0\end{pmatrix} \times \begin{pmatrix}1{,}5r\\ r\frac{\sqrt{3}}{2}\\ 0\end{pmatrix} = \begin{pmatrix}0\\ 0\\ -1{,}5r\omega\end{pmatrix}$$

$$\cancel{\bar v_B = \bar v_C + \bar\omega_{отн} \times \overline{CB};\quad -\bar v_C = \bar\omega_{отн} \times \overline{CB}}$$

$\omega_z = 0$

$$\cancel{\begin{pmatrix}0\\0\\1{,}5r\omega\end{pmatrix} = \begin{pmatrix}\omega_x\\ \omega_y\\ \omega_z\end{pmatrix} \times \begin{pmatrix}0{,}5r\\ -\sqrt{3}/2\, r\\ 0\end{pmatrix} = \begin{pmatrix}\omega_z \cdot \frac{\sqrt{3}}{2}\\ 0{,}5r\omega_z\\ -\omega_x r\frac{\sqrt{3}}{2} - \omega_y 0{,}5r\end{pmatrix}}$$

$$\cancel{-3r\omega = \omega_x r\sqrt{3} + \omega_y r;\quad -3\omega = \sqrt{3}\omega_x + \omega_y}$$

$$\cancel{9\omega^2 = 3\omega_x^2 + \omega_y^2 + 2\sqrt{3}\omega_x\omega_y}\quad \omega_y = -3\omega - \sqrt{3}\omega_x$$

$$|\bar v_C| = |\omega_{отн}| \cdot |CB| \sin\alpha;\quad 9\omega^2 = 4\omega_x^2 + 4\omega_y^2;\quad \cancel{9\omega^2 = 4\omega_x^2 + 9\omega^2 + 3\omega_x^2 + 6\sqrt{3}\omega\omega_x}$$

$$\cancel{7\omega_x = -6\sqrt{3}\omega;\quad \omega_x = -\frac{6\sqrt{3}}{7}\omega;\quad \omega_y = -3\omega + \sqrt{3}\frac{6\sqrt{3}}{7}\omega = -\frac{3}{7}\omega}$$

$$\cancel{\omega_K = \omega_{абс} = \omega_{пер} + \omega_{отн} = \bar\omega + \bar\omega_{отн} = \begin{pmatrix}0\\ \omega\\ 0\end{pmatrix} + \begin{pmatrix}-6\sqrt{3}/7\,\omega\\ -3/7\,\omega\\ 0\end{pmatrix} = \begin{pmatrix}-6\sqrt{3}/7\,\omega\\ 4/7\,\omega\\ 0\end{pmatrix}}$$

$$\cancel{\omega_K = \sqrt{\frac{16}{49}\omega^2 + \frac{108}{49}\omega^2} = \sqrt{\frac{124}{49}\omega^2} = \frac{2}{7}\omega\sqrt{31}}$$

$$\cancel{9\omega^2 = 4\omega_x^2 + 4(9\omega^2 + 6\sqrt{3}\omega_x\omega + 3\omega_x^2);\quad 27\omega^2 + 24\sqrt{3}\omega\omega_x + 16\omega_x^2 = 0}$$

$$\cancel{D = 1728\omega^2 - 1728\omega^2 = 0,\quad \omega_x = \frac{-24\sqrt{3}\omega}{32} = -\frac{3\sqrt{3}\omega}{4}}$$

$$\cancel{\omega_y = -3\omega - \sqrt{3} \cdot \frac{-3\sqrt{3}\omega}{4} = -3\omega + \frac{9\omega}{4} = -\frac{3\omega}{4}}$$

$$\cancel{\omega_K = \omega_{абс} = \omega_{пер} + \omega_{отн} = \bar\omega + \bar\omega_{отн} = \begin{pmatrix}0\\ \omega\\ 0\end{pmatrix} + \begin{pmatrix}-\frac{3\sqrt{3}\omega}{4}\\ -\frac{3\omega}{4}\\ 0\end{pmatrix} = \begin{pmatrix}-\frac{3\sqrt{3}\omega}{4}\\ 1/4\,\omega\\ 0\end{pmatrix}}$$

$$\cancel{\omega_K = \sqrt{\frac{\omega^2}{16} + \frac{27\omega^2}{16}} = \frac{\sqrt{7}}{2}\omega}$$

$$\bar v_C = \bar v_B + \bar\omega_K \times \overline{BC};\quad \bar\omega_K \times \overline{BC} = \bar\omega \times \overline{OC},\quad \omega_K BC \sin 90^\circ = \omega \cdot OC \sin 60^\circ$$

$$\frac{\sqrt{3}}{2}\ \omega_K \cdot r = \omega \cdot r\sqrt{3}\,\frac{\sqrt{3}}{2};\quad \underline{\omega_K = \omega\sqrt{3}};\quad \bar\omega_K = \omega_K \bar e_z$$

$$\bar\varepsilon_K = \frac{d\vec\omega_K}{dt} = \frac{d}{dt}\left(\omega_K \bar e_z + \cancel{\omega_K e_z}\right) = \frac{d\omega_K}{dt}\bar e_z + \omega_K \frac{d\bar e_z}{dt} =$$

$$= \sqrt{3}\,\varepsilon\,\bar e_z + \bar\omega_K \cdot \bar\omega = \sqrt{3}\,\bar\varepsilon + \bar\omega_K \times \bar\omega = \sqrt{3}\left(\bar\varepsilon + \omega\,\bar e_z \times \bar\omega\right) =$$

$\varepsilon_K = \sqrt{3\varepsilon^2 + 3\omega^4}$ $(\bar{\varepsilon}_r \perp \bar{\omega})$

$\bar{W}_B = \cancel{\bar{W}_O}^{0} + \bar{\varepsilon}_K \times \bar{OB} + \bar{\omega}_K \times [\bar{\omega}_K \times \bar{OB}] = \begin{pmatrix}\sqrt{3}\varepsilon\\0\\\sqrt{3}\omega^2\end{pmatrix} \times \begin{pmatrix}2r\\0\\0\end{pmatrix} + \begin{pmatrix}\omega\sqrt{3}\\0\\0\end{pmatrix} \times 0 =$

$= \begin{pmatrix}0\\+2\sqrt{3}r\omega^2\\0\end{pmatrix}$, $W_B = 2\sqrt{3}r\omega^2$; $\bar{BA} = (2r,0,0) - (r, r\sqrt{3}, 0) = (r, -r\sqrt{3}, 0)$

$W_a = \bar{W}_B + \bar{\varepsilon}_K \times \bar{BA} + \bar{\omega}_K \times [\bar{\omega}_K \cdot \bar{BA}] = \begin{pmatrix}0\\2\sqrt{3}r\omega^2\\0\end{pmatrix} + \begin{pmatrix}\sqrt{3}\varepsilon\\0\\\sqrt{3}\omega^2\end{pmatrix} \times \begin{pmatrix}r\\-r\sqrt{3}\\0\end{pmatrix} +$

$+ \begin{pmatrix}\sqrt{3}\omega\\0\\0\end{pmatrix} \times \left(\begin{pmatrix}\sqrt{3}\omega\\0\\0\end{pmatrix} \times \begin{pmatrix}r\\-r\sqrt{3}\\0\end{pmatrix}\right) = \begin{pmatrix}0\\2\sqrt{3}r\omega^2\\0\end{pmatrix} + \begin{pmatrix}+3r\omega^2\\-\sqrt{3}r\omega^2\\-3\varepsilon r\end{pmatrix} + \begin{pmatrix}0\\-3\sqrt{3}\omega^2 r\\0\end{pmatrix} =$

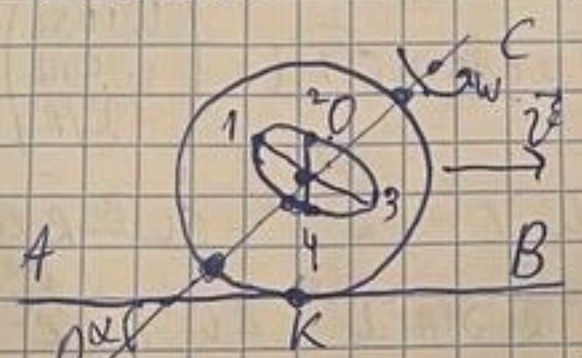
$= \begin{pmatrix}3r\omega^2\\-2\sqrt{3}r\omega^2\\-3\varepsilon r\end{pmatrix}$ $W_a = r\sqrt{9\varepsilon^2 + 21\omega^4}$

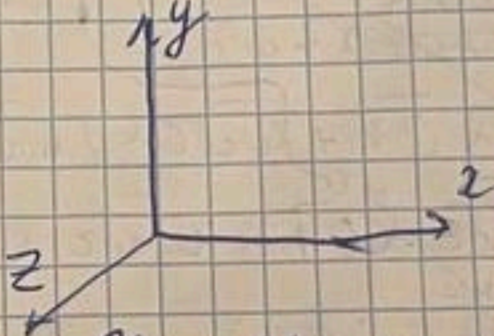
Ответ: $\omega_K = \omega\sqrt{3}$, $\varepsilon_K = \sqrt{3\varepsilon^2 + 3\omega^4}$; $W_a = r\sqrt{9\varepsilon^2 + 21\omega^4}$; $W_B = 2\sqrt{3}r\omega^2$

№4.12

Дано:
R, V
ω_0, r
α
v_1, v_2, v_3 - ?
W_1, W_2, W_3 - ?

Решение.





$v_O = v$, $v_K = 0$

$v_O = \begin{pmatrix}v\\0\\0\end{pmatrix}$

$\cancel{v_O} = v_O = \cancel{\bar{v}_K}^{0} + \bar{\omega}_0 \times \bar{KO}$; $\begin{pmatrix}v\\0\\0\end{pmatrix} = \begin{pmatrix}0\\0\\\omega\end{pmatrix} \times \begin{pmatrix}0\\+R\\0\end{pmatrix} = \begin{pmatrix}-\omega R\\0\\0\end{pmatrix}$

$\omega_0 = -\frac{v}{R}$

$\bar{v}_1 = \bar{v}_O + \bar{\omega}_{абс} \times \bar{O1}$, $\bar{O1} = \begin{pmatrix}-r\sin\alpha\\r\cos\alpha\\0\end{pmatrix}$

$\bar{\omega}_{абс} = \bar{\omega}_{пер} + \bar{\omega}_{отн} = \bar{\omega}_0 + \bar{\omega} =$

$= \begin{pmatrix}\omega\cos\alpha\\\omega\sin\alpha\\-\frac{v}{R}\end{pmatrix}$

$\cancel{\bar{v}_1 = \begin{pmatrix}v\\0\\0\end{pmatrix} + \begin{pmatrix}-r\sin\alpha\\r\cos\alpha\\0\end{pmatrix} \times \begin{pmatrix}\omega\cos\alpha\\\omega\sin\alpha\\-v/R\end{pmatrix} = \begin{pmatrix}v - v\frac{r}{R}\cos\alpha\\-v\sin\alpha\\-r\omega\end{pmatrix}}$

$\cancel{v_1 = \sqrt{v^2 + v^2\cos^2\alpha - 2v^2\cos\alpha + v^2\sin^2\alpha + r^2\omega^2} = \sqrt{2v^2(1-\cos\alpha) + r^2\omega^2}}$

$\bar{v}_1 = \begin{pmatrix}v\\0\\0\end{pmatrix} + \begin{pmatrix}\omega\cos\alpha\\\omega\sin\alpha\\-v/R\end{pmatrix} \times \begin{pmatrix}-r\sin\alpha\\r\cos\alpha\\0\end{pmatrix} = \begin{pmatrix}v(1+\frac{r}{R}\cos\alpha)\\-v\frac{r}{R}\sin\alpha\\\omega r\end{pmatrix}$ ← должен быть плюс почему-то

$v_1 = \sqrt{v^2(1+\frac{r}{R}\cos\alpha)^2 + v^2\frac{r^2}{R^2}\sin^2\alpha + \omega^2 r^2}$; $\bar{O2} = (0, 0, -r)^T$

$\bar{v}_2 = \bar{v}_O + \bar{\omega}_{абс} \times \bar{O2} = \begin{pmatrix}v\\0\\0\end{pmatrix} + \begin{pmatrix}\omega\cos\alpha\\\omega\sin\alpha\\-v/R\end{pmatrix} \times \begin{pmatrix}0\\0\\-r\end{pmatrix} = \begin{pmatrix}v - \omega\sin\alpha r\\-\omega\cos\alpha r\\0\end{pmatrix}$

$v_2 = \sqrt{v^2 + \omega^2 r^2 - 2v\omega\sin\alpha r}$

$$\bar{v}_3 = \bar{v}_0 + \bar{\omega}_{абс} \times \bar{O3}, \quad \bar{O3} = \begin{pmatrix} r\sin\alpha \\ -r\cos\alpha \\ 0 \end{pmatrix}$$

$$\varepsilon_{абс} = \frac{d\bar{\omega}_{абс}}{dt} = \frac{d}{dt}(\omega_{пер} + \omega_{отн}) = \frac{d}{dt}(\bar{\omega} + \bar{\omega}_0) = \frac{d}{dt}(\omega \bar{e} + \omega_0 \bar{e}_0) = \omega \cdot \frac{d\bar{e}}{dt} + \omega_0 \cdot \frac{d\bar{e}_0}{dt} = \omega_0 \frac{d\bar{e}_0}{dt} = \bar{\omega} \times \bar{\omega}_0$$

$$\bar{v}_3 = \begin{pmatrix} v \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} \omega\cos\alpha \\ \omega\sin\alpha \\ -v/R \end{pmatrix} \times \begin{pmatrix} r\sin\alpha \\ -r\cos\alpha \\ 0 \end{pmatrix} = \begin{pmatrix} v - v\cos\alpha \frac{r}{R} \\ v\sin\alpha \frac{r}{R} \\ -\omega r \end{pmatrix}$$

$\cancel{v_3 = v - v\cos\alpha}$

a

$$v_3 = \sqrt{v^2\left(1 + \frac{r^2}{R^2}\right) + \omega^2 r^2 - 2v^2\cos\alpha \frac{r}{R}}$$

$$\cancel{\bar{W}_1 = \bar{W}_0 + \varepsilon_{абс} \times \bar{O1}} + [\bar{\omega}_{абс} \times [\bar{\omega}_{абс} \times \bar{O2}]] = \begin{pmatrix} \omega\cos\alpha \\ \omega\sin\alpha \\ -v/R \end{pmatrix} \times \begin{pmatrix} -\omega\sin\alpha\, r \\ -\omega\cos\alpha\, r \\ 0 \end{pmatrix} =$$

$$= \begin{pmatrix} v\omega\cos\alpha\, r/R \\ -v\omega\sin\alpha\, r/R \\ -\omega^2 r \end{pmatrix}, \quad \cancel{W_1 = \sqrt{\frac{v^2}{\omega^2}\frac{r^2}{R^2} + \omega^4 r^2}}$$

$$\bar{W}_1 = \bar{W}_0 + \cancel{\varepsilon_{абс} \times \bar{O1} + \bar{\omega}_{абс} \times [\bar{\omega}_{абс} \times \bar{O1}]} = [\bar{\omega} \times \bar{\omega}_0] \times \bar{O1} + \begin{pmatrix} v\omega\cos\alpha \frac{r}{R} \\ -v\omega\sin\alpha\, r/R \\ -\omega^2 r \end{pmatrix} =$$

$$= \begin{pmatrix} \omega\cos\alpha \\ \omega\sin\alpha \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -v/R \end{pmatrix} \times \begin{pmatrix} -r\sin\alpha \\ r\cos\alpha \\ 0 \end{pmatrix} + \begin{pmatrix} \\ \\ \end{pmatrix} = \begin{pmatrix} -\omega\sin\alpha \frac{v}{R} \\ \omega\cos\alpha\, v/R \\ 0 \end{pmatrix} \times \begin{pmatrix} -r\sin\alpha \\ r\cos\alpha \\ 0 \end{pmatrix} + \begin{pmatrix} \\ \\ \end{pmatrix} =$$

$$= \begin{pmatrix} v\omega\cos\alpha \frac{r}{R} \\ -v\omega\sin\alpha\, r/R \\ \cancel{\omega r \frac{v}{R}\sin\alpha\cos\alpha} + \omega^2 r \end{pmatrix} \quad W_1 = \sqrt{v^2\omega^2 \frac{r^2}{R^2} + \omega^4 r^2} = v\omega r\sqrt{v^2/R^2 + \omega^2}$$

должен быть плюс

?

$$W_3 = \varepsilon_{абс} \times \bar{O3} + [\bar{\omega}_{абс} \times [\bar{\omega}_{абс} \times \bar{O3}]] =$$

$$\bar{W}_1 = \varepsilon_{абс} \times \bar{O1} + \bar{\omega}_{абс} \times [\bar{\omega}_{абс} \times \bar{O1}] = \begin{pmatrix} \omega\cos\alpha \\ \omega\sin\alpha \\ -v/R \end{pmatrix} \times \begin{pmatrix} v\frac{r}{R}\cos\alpha \\ -v\, r/R\sin\alpha \\ \omega r \end{pmatrix} =$$

$$= \begin{pmatrix} \omega^2 r\sin\alpha + \frac{v^2}{R^2} r\sin\alpha \\ \omega^2 r\cos\alpha + \frac{v^2}{R^2} r\cos\alpha \\ \cancel{-2\omega v\, r/R\cos\alpha\sin\alpha}\ 0 \end{pmatrix}, \quad W_1^2 = \omega^4 r^2 + \frac{v^2}{R^2} r^2 - \cancel{2\omega v \frac{r}{R}\cos\alpha\sin\alpha} - $$

$$+ 2\omega^2 r^2 \frac{v^2}{R^2}\sin^2\alpha + 2\omega^2 r^2 \frac{v^2}{R^2}\cos^2\alpha$$

$$W_1 = \omega^2 r + \frac{v^2}{R^2} r$$

$W_3 = W_1$ (в скобках поменяются знаки, сумма будет та же)

$$W_2 = \bar{\omega}_{абс} \times [\bar{\omega}_{абс} \times \bar{O2}] = \begin{pmatrix} \omega\cos\alpha \\ \omega\sin\alpha \\ -v/R \end{pmatrix} \times \begin{pmatrix} -\omega\sin\alpha\, r \\ -\omega\cos\alpha\, r \\ 0 \end{pmatrix} = \begin{pmatrix} v\omega\cos\alpha\, r/R \\ -v\omega\sin\alpha\, r/R \\ -\omega^2 r \end{pmatrix}$$

$$\cancel{W_2 = \sqrt{\frac{v^2}{\omega^2}\frac{r^2}{R^2} + \omega^4 r^2}}; \quad \varepsilon_{абс} \times \bar{O2} = \begin{pmatrix} -\omega\sin\alpha \frac{v}{R} \\ \omega\cos\alpha\, v/R \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -r \end{pmatrix} = \begin{pmatrix} \omega\sin\alpha \frac{vr}{R} \\ \omega\cos\alpha \frac{vr}{R} \\ 0 \end{pmatrix}$$

$$W_2 = \sqrt{\omega^4 r^2 + v^2\omega^2 \frac{r^2}{R^2}(\cos\alpha + \sin\alpha)^2 + v^2\omega^2\frac{r^2}{R^2}(\sin\alpha - \cos\alpha)^2} =$$

$$= \sqrt{\omega^4 r^2 + 2v^2\omega^2 \frac{r^2}{R^2}} = W_4$$

Ответ: $v_1 = \sqrt{v^2\left(1 + \frac{r}{R}\cos\alpha\right)^2 + v^2\frac{r^2}{R^2}\sin^2\alpha + \omega^2 r^2}$, $v_2 = \sqrt{v^2 + \omega^2 r^2 - 2v\sin\alpha\, r\omega}$

$v_3 = \sqrt{v^2\left(1 + \frac{r}{R}\right)^2 + \omega^2 r^2 - 2v^2\cos\alpha \frac{r}{R}}$, $W_1 = W_3 = \omega^2 r + \frac{v^2}{R^2} r$

$W_2 = \sqrt{\omega^4 r^2 + 2v^2\omega^2 \frac{r^2}{R^2}}$

№4.23

Дано:
$x_1 - W_1$
$x_2 - W_2$
$x, W - ?$

Решение:

$\bar{W_x} = \bar{W_{x_1}} + \bar{\varepsilon} \times \overline{x_1 x} + \bar{\omega} \times [\bar{\omega} \times \overline{x_1 x}] = \bar{W_{x_1}} + \bar{\varepsilon} \times (\bar{x} - \bar{x_1}) + \omega^2 (\bar{x} - \bar{x_1})$

$\bar{W_x} = \bar{W_{x_2}} + \bar{\varepsilon} \times \overline{x_2 x} + \bar{\omega} \times [\bar{\omega} \times \overline{x_2 x}] = \bar{W_{x_2}} + \bar{\varepsilon} \times (\bar{x} - \bar{x_2}) + \omega^2 (\bar{x} - \bar{x_2})$

$\bar{W_{x_2}} = \bar{W_{x_1}} + \bar{\varepsilon} \times \overline{x_1 x_2} + \bar{\omega} \times [\bar{\omega} + \overline{x_1 x_2}] = \bar{W_{x_1}} + \bar{\varepsilon} (\bar{x_2} - \bar{x_1}) + \omega^2 (\bar{x_2} - \bar{x_1})$

$\bar{W_x} = \bar{W_{x_1}} + \bar{a}(x - x_1) + \bar{b}(x - x_1) = \bar{W_{x_1}} + (x - x_1)(\bar{a} + \bar{b})$

$\bar{W_x} = \bar{W_{x_2}} + (x - x_2)(\bar{a} + \bar{b})$, $\bar{W_{x_2}} = \bar{W_{x_1}} + (x_2 - x_1)(\bar{a} + \bar{b})$

$\bar{a} + \bar{b} = \dfrac{\bar{W_{x_2}} - \bar{W_{x_1}}}{x_1 - x_2}$; $\bar{W_x} = \bar{W_{x_1}} + (x - x_1) \cdot \dfrac{\bar{W_{x_2}} - \bar{W_{x_1}}}{x_2 - x_1} =$

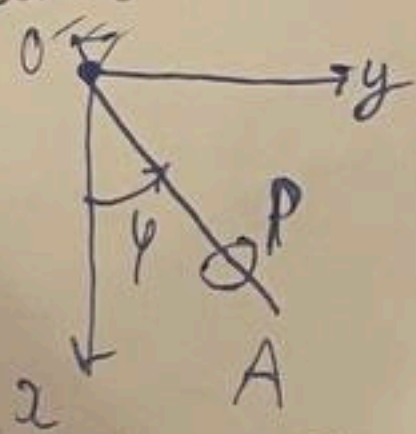
$= \bar{W_{x_1}} \cdot \left(1 - \dfrac{x - x_1}{x_2 - x_1}\right) + \bar{W_{x_2}} \cdot \dfrac{x - x_1}{x_2 - x_1} = \dfrac{x_2 - x}{x_2 - x_1} \bar{W_1} + \dfrac{x - x_1}{x_2 - x_1} \bar{W_2}$

Ответ: $\dfrac{x_2 - x}{x_2 - x_1} \bar{W_1} + \dfrac{x - x_1}{x_2 - x_1} \bar{W_2}$

№2.16

Дано:
$\varphi = \varphi_0 \sin \omega t$
$OP = at^2 \cdot \frac{1}{2}$
$v = ?$
$W = ?$

Решение:



~~$v_r = r'$, $v_\varphi = \varphi' r$~~
~~$v_r = OP' = at$~~
~~$v_\varphi = \omega \varphi_0 \sin \cos \omega t =$~~

$\bar{v}_{абс} = \bar{v}_{отн} + \bar{v}_{пер}$, $v_{отн} \perp v_{пер}$

$v_{отн} = \dfrac{dr}{dt} = at$, $v_{пер} = r \cdot \dfrac{d\bar{e}}{dt} = r \cdot \dot{\varphi} =$

$= \frac{1}{2} at^2 \cdot \omega \varphi_0 \cos(\omega t)$.

$v = \sqrt{a^2 t^2 + \frac{1}{4} a^2 t^4 \omega^2 \varphi_0^2 \cos^2(\omega t)} = \dfrac{at}{2} \sqrt{4 + \omega^2 t^2 \varphi_0^2 \cos^2(\omega t)}$

$\bar{W}_{абс} = W_{отн} + W_{пер} + W_{кор}$; ~~$W_{отн} = a$, $W_{пер} = r\ddot{\varphi} = -\frac{1}{2} at^2 \omega^2 \varphi_0 \sin \omega t$~~

~~$W_{кор} = 2\omega v_{отн} = 2\omega a t$~~ ~~$W_{кор} = 2\omega v_{отн}$~~

~~$W_{абс} = \sqrt{4\omega^2 a^2 t^2 + a^2 + \frac{1}{4} a^2 t^4 \omega^4 \varphi_0^2 \sin^2 \omega t}$~~ $= a\sqrt{4\omega^2 t^2 + 1 + \frac{1}{4} t^4 \varphi_0^2 \omega^4 \sin^2 \omega t}$

$W_r = \cancel{a} \; \ddot{r} - r\dot{\varphi}^2 = a - \frac{1}{2} a t^2 \omega^2 \varphi_0^2 \cos^2(\omega t)$

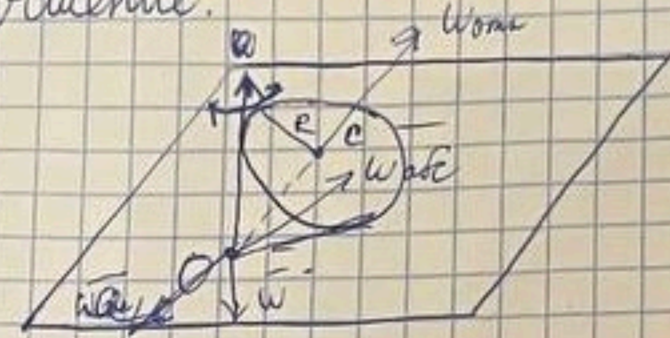
$W_\varphi = r\ddot{\varphi} + 2\dot{r}\dot{\varphi} = -\frac{1}{2} a t^2 \omega^2 \varphi_0 \sin \omega t + 2at\omega \varphi_0 \cos \omega t$

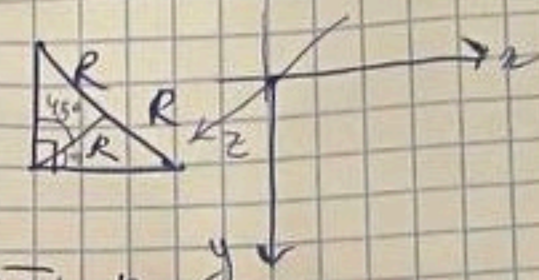
$W = \sqrt{W_r^2 + W_\varphi^2} = a\sqrt{\left(1 - \frac{1}{2} t^2 \omega^2 \varphi_0^2 \cos^2 \omega t\right)^2 + \left(2t\omega\varphi_0 \cos \omega t - \frac{1}{2} t^2 \omega^2 \varphi_0 \sin \omega t\right)^2}$

№4.34

Дано:
R
$v_C = v$
W_C - ?

Решение:





$|OC| = R$

$\bar v_m = \bar v_{отн_m} + \bar v_{пер_m} = \bar v + \bar v_{пер_m}$; $v_{пер\,m} = \bar v_C = \cancel{\bar v_0}^{\,0} + \bar\omega \times \bar{OC}$

$\bar v_{отн\,m} \perp \bar v_C$

$\cancel{\bar W_m = (\bar v_{пер})' = (\bar\omega_{абс} \times \bar{OC})' = \frac{d\omega_{абс}}{dt} \times \bar{OC} + \bar\omega \times \frac{d\bar{OC}}{dt} =}$

$\cancel{= \frac{d\bar\omega_e}{dt} \times \bar{OC} = = \frac{d\omega_{отн}}{dt} \times \bar{OC} =}$

$\cancel{\bar W_m}$ $\bar W_m = \bar W_C^{пер} + \bar W_m^{отн} + \bar W_{кор} = \bar W_0 + \bar\varepsilon \times \bar{OC} + \bar\omega \times [\bar\omega \times \bar a] +$
$+ 0 + 2\bar\omega \times \bar v_{отн} = \bar\omega \times [\bar\omega \times \bar{OC}] + 2\bar\omega \times \bar v_{отн}$

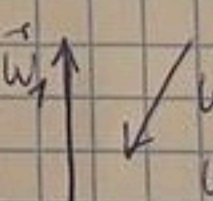
$\bar v_C = \bar\omega \times OC$; $\begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ \omega \\ 0 \end{pmatrix} \times \begin{pmatrix} R\frac{\sqrt2}{2} \\ -R\cdot\sqrt2/4 \\ 0 \end{pmatrix}$; $v = -\omega R \frac{\sqrt2}{4}$

$\omega = -\sqrt2 \frac{v}{R}$

$\bar W_m = \bar\omega \times \bar v_C + 2\bar\omega \times \bar v_{отн} = \begin{pmatrix} 0 \\ -\sqrt2\, v/R \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix} + \begin{pmatrix} 0 \\ -2\sqrt2 \frac{v}{R} \\ 0 \end{pmatrix} \times \begin{pmatrix} v\frac{\sqrt2}{2} \\ -v\frac{\sqrt2}{2} \\ 0 \end{pmatrix} =$

$= \begin{pmatrix} -\sqrt2\, v^2/R \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 2v^2/R \end{pmatrix} = \begin{pmatrix} -\sqrt2\, v^2/R \\ 0 \\ 2v^2/R \end{pmatrix}$; $W_m = \sqrt{4v^4/R^2 + 2v^2/R} =$
$= \sqrt6 \frac{v^2}{R}$

Ответ: $\sqrt6 \frac{v^2}{R}$

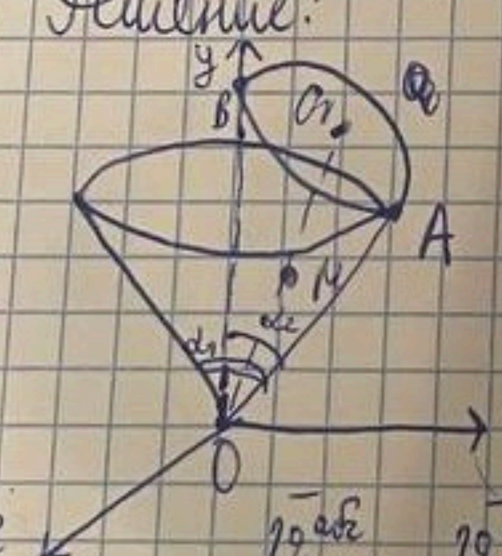


$\omega_1 = + \frac{v}{l \sin \alpha_2/2}$

№13

Дано:
$\alpha_2 = 45°$
$\alpha_1 = 90°$
$OO_1 = l$
$v_{O_1} = v$
$OM = f(t)$
$OM = MO_1$
v_m, W_m - ?

Решение:



$\bar v_M^{абс} = \bar v_M^{отн} + \bar v_M^{пер}$

$|\bar v_M^{отн}| = f'(t)$

$\bar v_M^{пер} = \cancel{\bar v_0}^{\,0} + \bar\omega_1 \times \bar{OM}$

$\bar v_M^{пер} = \begin{pmatrix} 0 \\ +v/l \sin\frac{\alpha_2}{2} \\ 0 \end{pmatrix} \times \begin{pmatrix} \frac{l}{2}\sin\frac{\alpha_2}{2} \\ \frac{l}{2}\cos\frac{\alpha_2}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -\frac{v}{2} \end{pmatrix}$

$\bar v_{O_1} = \cancel{\bar v_0}^{\,0} + \bar\omega_1 \bar{OO_1}$

$\begin{pmatrix} 0 \\ 0 \\ -v \end{pmatrix} = \begin{pmatrix} 0 \\ \omega_1 \\ 0 \end{pmatrix} \times \begin{pmatrix} l \sin\alpha_2/2 \\ l\cos\frac{\alpha_2}{2} \\ 0 \end{pmatrix}$

$\bar v_M^{абс} = \bar v_M^{отн} + \bar v_M^{пер} = \begin{pmatrix} f'(t)\sin\frac{\alpha_2}{2} \\ f'(t)\cos\frac{\alpha_2}{2} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ -v/2 \end{pmatrix} = \begin{pmatrix} f'(t)\sin\frac{\alpha_2}{2} \\ f'(t)\cos\frac{\alpha_2}{2} \\ -v/2 \end{pmatrix}$

$v_{M}^{абс} = \sqrt{f'(t)^2 + \frac{v^2}{4}}$

$\bar W_M^{абс} = \bar W_M^{отн} + \bar W_{кор} + \bar W_M^{пер}$; $\bar W_M^{отн} = \begin{pmatrix} f''(t)\sin\frac{\alpha_2}{2} \\ f''(t)\cos\frac{\alpha_2}{2} \\ 0 \end{pmatrix}$; $\bar W_M^{кор} = 2\bar\omega_1 \times \bar v_{отн\,M} =$

$= \begin{pmatrix} 0 \\ 2v/l\sin\frac{\alpha_2}{2} \\ 0 \end{pmatrix} \times \begin{pmatrix} f'(t)\sin\frac{\alpha_2}{2} \\ f'(t)\cos\alpha_2/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -\frac{2v}{l} f'(t) \end{pmatrix}$ $\bar W_M^{пер} = \cancel{\bar W_0}^{\,0} + \cancel{\varepsilon \times \bar{OM}}^{\,0} + \bar\omega_1 \times [\bar\omega_1 \times \bar{OM}] = \begin{pmatrix} 0 \\ v/l\sin\frac{\alpha_2}{2} \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -v/2 \end{pmatrix} =$ $\begin{pmatrix} -\frac{v^2}{l\sin\frac{\alpha_2}{2}} \\ 0 \\ 0 \end{pmatrix}$ $\Rightarrow W_{абс\,M} = \Rightarrow$ первая страница